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Paleomagnetic evidence for the clockwise rotation of Southwest Japan

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Over 500 oriented samples of felsic rocks of Cretaceous to Middle Miocene age were collected along the Gō River in the central part of Southwest Japan, in an attempt to detect the process of tectonic rotation of Southwest Japan from the paleomagnetic view point. Thermal demagnetization was successful in isolating characteristic directions from the remanent magnetization of samples. Reliability of the paleomagnetic direction is ascertained through the agreement of directions from different kinds of rocks as well as the presence of both normal and reversed polarities. The paleomagnetic results establish that Southwest Japan began to rotate clockwise through 58 ± 14° later than 28 Ma and ceased its motion by about 12 Ma. Southwest Japan has undergone no detectable north-south translation since 28 Ma. These results imply that southwest Japan was rotated about the pivot around 34°N, 129°E between 28 Ma and 12 Ma in association with the opening of the Japan Sea.

1. Introduction

Paleomagnetic studies of Cretaceous rocks of Southwest Japan [1,2] have established that the directions of Cretaceous sedimentary and volcanic rocks have shown clockwise deflection from the expected paleofield direction. This anomalous direction is explained by a clockwise rotation of Southwest Japan associated with the opening of the Japan sea [3,4]. Similar easterly declination data are reported from Paleogene rocks [5]. These facts leave little doubt that the clockwise rotation mainly occurred during the latter part of the Tertiary. The aim of the present work is to measure the process of clockwise rotation of Southwest Japan by paleomagnetism in an attempt to assess the timing and the amount of the rotation. The paleomagnetic results shed some light on the origin of the Japan Sea.

Felsic igneous rocks of Late Cretaceous to Miocene age are common over large areas of the Inner Zone of Southwest Japan. These rocks are ideal for studying the tectonic rotation of Southwest Japan, because these rocks of Cretaceous and Paleogene age have easterly characteristic directions of Southwest Japan [1,5]. These igneous rocks are extensively exposed along the Gō River and the neighbouring region in the central part of Southwest Japan, where this study was carried out (Fig. 1). This area has been geologically mapped in detail, and its stratigraphic relations have been well studied [6,7]: The igneous rocks of the area consist of a sequence of pyroclastic flow deposits and lava flows, which are intruded by plutonic rocks and dykes. Dacitic to rhyolitic compositions predominate between the Cretaceous and the Paleogene. During the Miocene, the composition changes to intermediate. Recently chronological studies by Matsuda [8] using fission-track methods have greatly clarified the history of igneous activity in this area which was originally worked out by Kawano and Ueda [9] using the K-Ar dating

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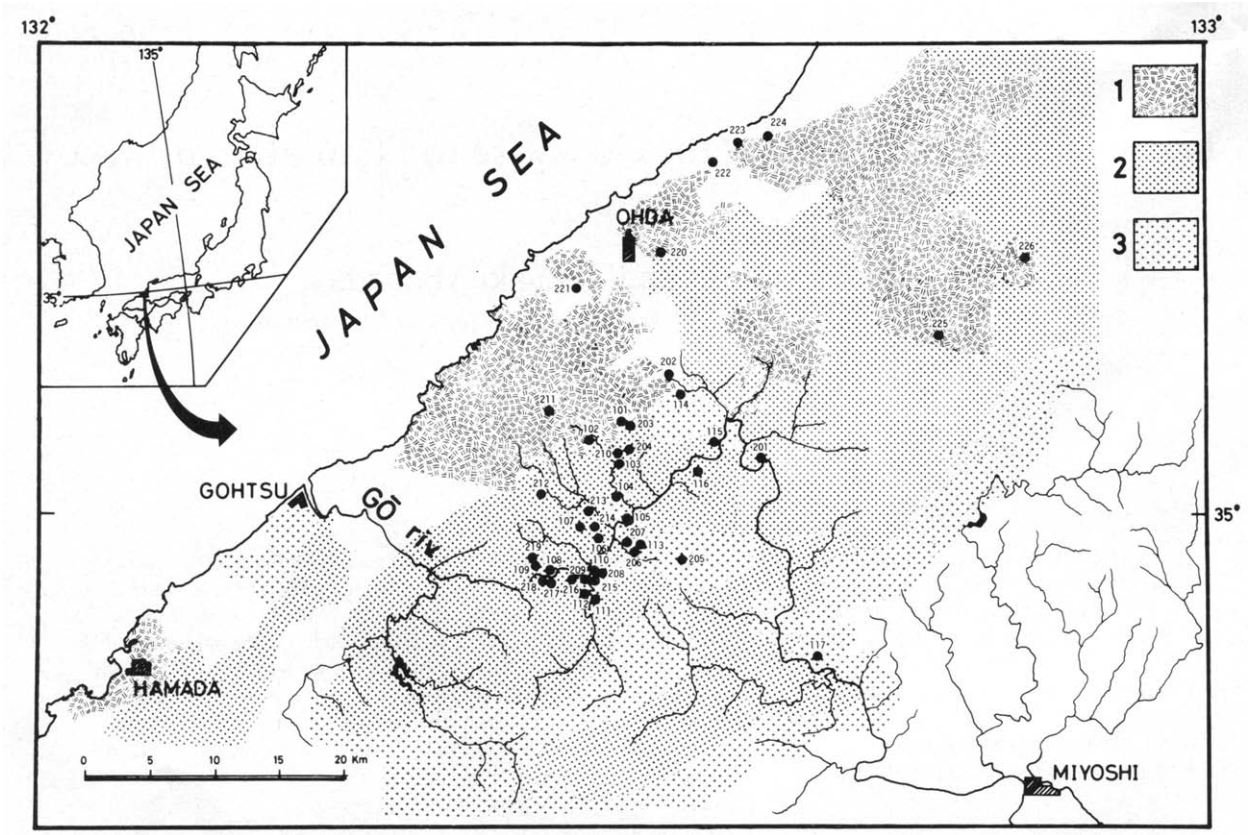


Fig. 1. Map showing the sampling localities together with distribution of Cretaceous to Tertiary felsic rocks. *1* = Neogene rocks; *2* = Paleogene rocks; *3* = Cretaceous rocks.

method. Seventeen radiometric ages have been obtained.

The bedding planes of strata of this area are clearly recognized from the intercalated thin layer of fine tuffaceous sediments and eutaxitic structure of the granules of pumice. The declination change with time is traced back from the Miocene to the Cretaceous precisely enough to obtain the process of rotational motion of Southwest Japan.

2. Sampling and measurements

More than 500 samples were collected by hand sampling oriented blocks from 45 sites using a magnetic compass. These sites included all the sampling sites for fission-track dating. Samples cover Cretaceous to Miocene in age. Sampling sites are shown in Fig. 1. Each site typically comprises twelve independently oriented samples distributed over distances ranging up to 20 m. Dips of the strata at all sites were observed in the field except for plutonic rocks. Uncertainty of a few degrees was introduced in determining the dip. Plutonic rocks were collected from several different stocks with the same age which were distributed in the studied area. Errors caused by tilting of the stocks probably partly cancels out after calculation of a formation mean for the granitic rocks, because these stocks are expected to be subjected to independent tilting movements.

Two cylindrical specimens were drilled from each oriented hand sample in the laboratory. Remanent magnetization was measured with a Schonstedt SSM-1A spinner magnetometer. Three specimens from each site were progressively

demagnetized in alternating field (AF) in steps of 50 Oe up to a maximum field of 500 Oe. Progressive thermal demagnetization was also undertaken on three specimens from each site in steps of 100° or 150°C to the Curie temperature with the increment steps reduced to 30° or 50°C near the Curie temperature. The AF demagnetizer has a three-axis tumbler, and ambient field cancellation by a Helmholtz coil system. The thermal demagnetizer has three cylindrical mu metal shields to provide an ambient field of about 10 gamma.

3. Results

3.1. Demagnetization

Treatment in progressive AF and thermal demagnetization for three pilot specimens from each site divided the sites into three categories (see Fig. 2): (A) natural remanent magnetization (NRM) shows little change in direction with either AF or thermal treatment (Fig. 2A); (B) NRM directions moved after each thermal treatment step and ultimately converged about a mean direction which was assumed to be that of the primary magnetization, but directions were quite stable under AF treatment (Fig. 2B); and (C) NRM directions moved toward a direction of a primary magnetization with both treatments (Fig. 2C); in general thermal treatment was more successful in isolating a coherent magnetization than AF treatment.

The optimum demagnetization temperature to produce minimum dispersion was selected from three pilot specimens for each site of category A, and the other specimens from the site were demagnetized at this temperature. All specimens in the other categories were progressively treated with thermal demagnetization so that the primary component was recognized in individual specimens. The mean direction at the demagnetization level yielding the highest value of Fisher precision parameter, k, [10] was generally accepted to be that of the primary magnetization.

Samples of rhyolite had stable magnetizations after thermal demagnetization above 400°C and their directions were tightly clustered. All samples

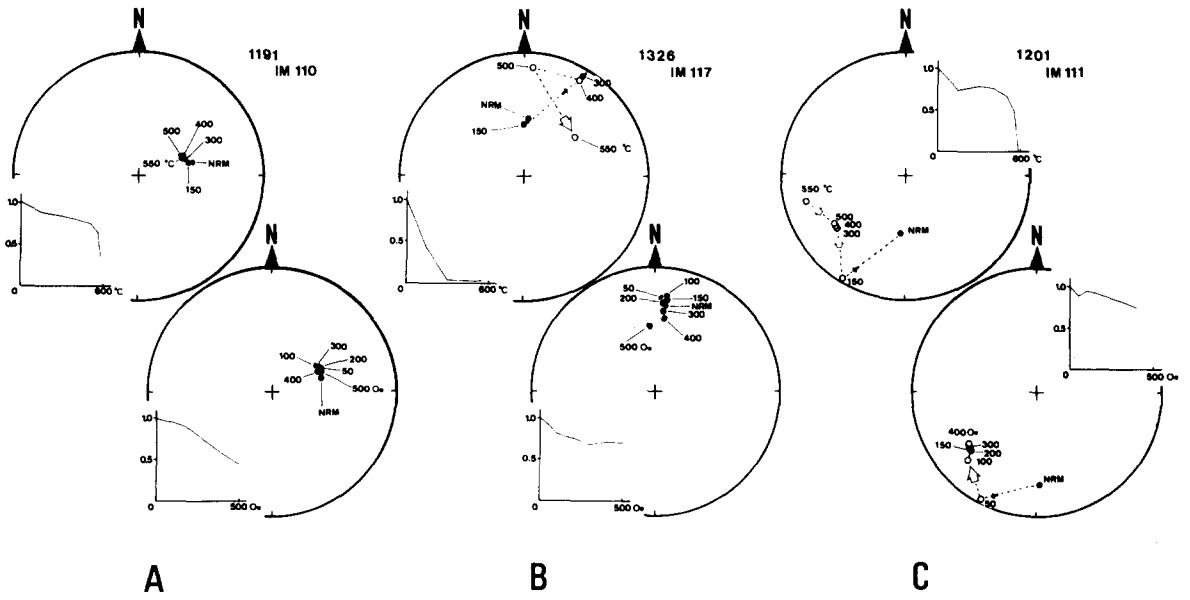


Fig. 2. Demagnetization data for specimens in the three categories A, B and C (see text). Thermal demagnetization steps are indicated by the temperature in °C while alternating field demagnetization steps are indicated by the value of the peak field in Oe. Projections are equal area, solid symbols on the lower hemisphere, open symbols on the upper hemisphere.

sustained only a very small component of magnetization after thermal demagnetization to 600°C (Fig. 2C). This indicates that the remanent magnetization resides in magnetite rather than hematite. The NRM directions of four sites of these rocks were highly divergent ($50° < \alpha_{95}$), although each specimen showed a high stability in response to both AF and thermal demagnetizations. Random directions are probably due to either the deformation of the sections at these sites or to inadvertent sampling of slumped blocks.

The magnetization superimposed on the primary component could be successfully removed by thermal demagnetization. Samples of the granite or granodiorite represent a typical example (Fig. 3). NRM directions of altered samples showed divergent distribution and their mean directions were significantly different from those of fresh samples from the same rock unit. During the thermal treatment steps the directions moved toward the stable direction for fresh samples, and formed a tight cluster after demagnetization at about 500°C (Fig. 3). Secondary magnetization induced during the process of preparing cylindrical specimens, parallel to the drill axis [11], was also successfully erased by thermal treatment at about 500°C.

Loss of specimens occurred during thermal demagnetization, as some specimens exploded at about 400°C. The following criteria were adopted to determine the reliability of each site: an acceptable site has at least five specimens cut from individual oriented samples and a mean direction

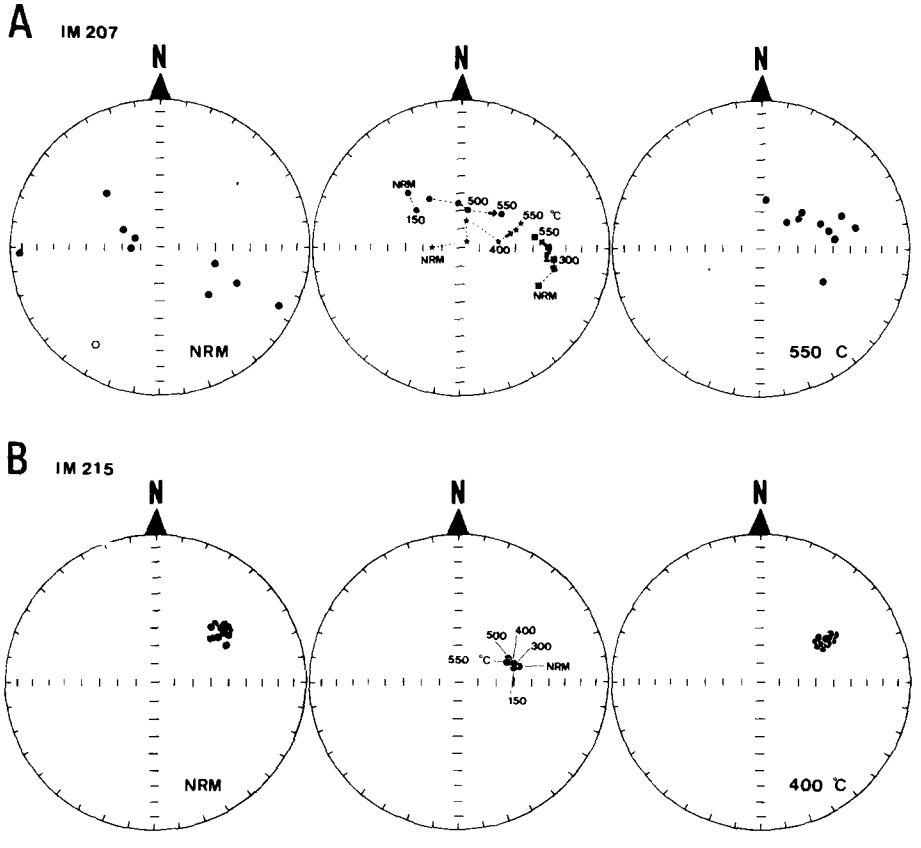


Fig. 3. Comparison of behavior of NRM in response to thermal demagnetization between the altered granitic sample (A) and the fresh one (B). Left: NRM directions before demagnetization of all samples of sites 207 and 215. Center: Progressive thermal demagnetization. Right: NRM directions of all samples after thermal demagnetization at 500°C and 400°C for sites 207 and 215, respectively. Projections are equal area, solid symbols on the lower hemisphere, open symbols on the upper hemisphere.

from them has a mean circle of 95% confidence of less than 20°.

3.2. Characteristic palaeomagnetic direction

Thirty five sites survived the criteria. The paleomagnetic data and the categories of magnetic behaviour of samples during demagnetization of each site are listed in Table 1, corrected for geological dip.

The reliable paleomagnetic data cover the age between 92 and 12 Ma. Radiometric age indicates that the paleomagnetic data can be divided into four stages; 80–92 Ma, 58–63 Ma, 28–33 Ma and 12–23 Ma. The cleaned paleomagnetic directions of each stage are shown in Fig. 4.

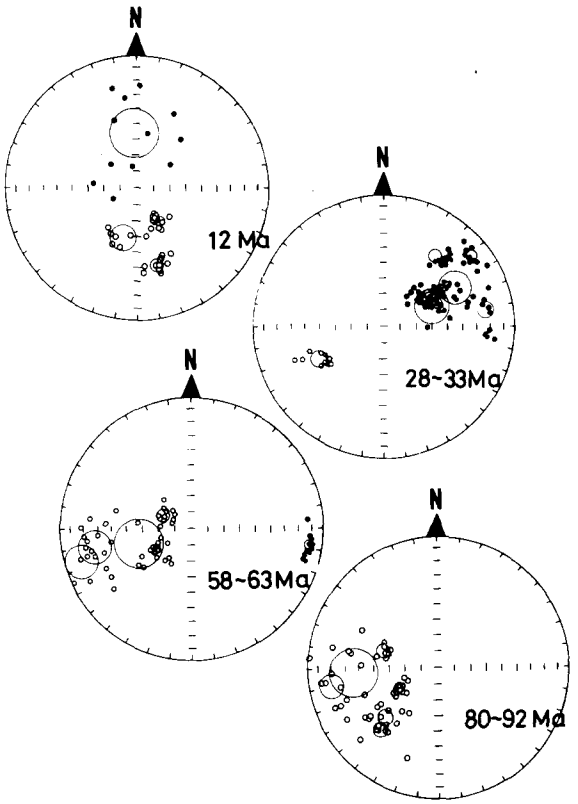


Fig. 4. Summary of characteristic directions of magnetization of samples with site mean directions and α_{95} confidence circles for four periods: 80–92 Ma (Cretaceous), 58–63 Ma (Paleocene), 28–33 Ma (Oligocene) and 12 Ma (Miocene). Projections are equal area, solid symbols on the lower hemisphere, open symbols on the upper hemisphere.

80–92 Ma (Late Cretaceous). Samples from all sites have a reversed polarity. Their directions show a clockwise deflection of the declination by more than 70°. The reliability of this direction is ascertained by agreement between directions of sediments and tuffs, although there is only a single polarity in magnetization. All the samples may have acquired their magnetization during the reversed interval between magnetic anomalies 33 and 34 [12].

58–63 Ma (Paleocene). The declination is 81 ± 26° with 95% confidence. The reliability of the direction is confirmed by the presence of both normal and reversed polarities as well as high agreement between data from lavas and tuffs.

28–33 Ma (Oligocene). The direction from granodiorite (29 Ma and 33 Ma) and from tuff (28 Ma) show a high degree of consistency. Evidence from eleven sites shows that the clockwise deflection of declination still persisted in the Late Oligocene.

At sites of the Sorajo Formation (sites 103, 210, 204; 29 Ma) a partial field reversal is apparently recorded. While these sites have anomalous westerly declination (by about 90°), inclination changes systematically: the younger sites have more steeply downward inclinations than the older site. The NRM of a ring dyke (site 104) may be acquired during this field reversal. These directions were excluded from the statistical computation for this stage.

12–23 Ma (Miocene). The paleomagnetic direction of the Ōmori Formation is almost the same as the axial dipole field. The Ōmori Formation is assigned to the Middle Miocene (about 12 Ma) from the geological correlation with the Kuri Formation of 14 Ma [13,14].

The declination change probably occurred during the Early Miocene. Data from two sites, 220 (14 Ma) and 202 (23 Ma), may show characteristic directions acquired during the period of declination change. The NRMs of two other sites (203, 226; 23 Ma) formed during this period are coincident with the direction of the present magnetic field, although the gabbroic rocks show a reversed polarity. It is possible that the primary component

TABLE 1

Paleomagnetic results for the Gō River region

Geological sequence		Age (Ma)	Site	Category	DT (°C)	N	D(°)	I(°)	α_{95} (°)	k	Rock type
Izumo Group	Ōmori Fm.	(12) [a]	221	C	500	8	−165.2	−58.7	8.7	41.8	andesite lava
			222	A	400	12	164.5	−39.6	3.0	210.3	andesite lava
			223*	A	400	12	148.5	−65.9	2.8	242.9	dacite welded tuff
			224*	C	400	12	−3.0	57.2	16.6	7.8	andesite lava
Iwami Group	Kuri Fm.	(14) [a]	220	C	500	12	25.0	50.9	12.0	14.0	dacite lava
Hata Subgroup		23	202*	C	550	9	−104.2	−64.1	6.1	72.3	andesite lava
			203*	C	550	12	−2.3	44.8	5.2	69.5	andesite lava
		23	226*	A	500	11	−177.5	−52.1	2.7	283.3	gabbro
Kawauchi Group	Ringdyke		104*	A	500	10	−118.6	30.5	5.5	77.4	rhyodacite dyke
	Sorajo Fm.	29	103*	C	500	10	−83.5	64.5	9.8	25.1	dacite welded tuff
			210*	C	500	11	−101.8	52.1	3.6	165.6	dacite welded tuff
			204*	C	500	11	−96.1	30.5	11.2	17.2	dacite welded tuff
	Sasahara Fm.	28	107*	A	400	10	52.4	15.9	3.6	185.6	dacite welded tuff
			212*	A	500	11	83.5	21.6	5.8	63.2	dacite welded tuff
			213*	A	500	12	42.0	30.9	1.8	552.1	dacite welded tuff
			214*	A	500	11	35.6	35.1	3.6	164.1	dacite welded tuff

TABLE 1 (continued)

Geological sequence		Age (Ma)	Site	Category	DT (°C)	N	D(°)	I(°)	α_{95} (°)	k	Rock type
Kawamoto Granodiorite		29 (25) [b]	105	C	400	9	61.4	39.5	11.5	21.1	granodiorite
		33	110	A	400	8	64.3	55.4	3.2	305.0	granodiorite
			207	C	550	10	68.3	57.5	11.0	20.3	granodiorite
			208	C	400	5	59.6	56.8	5.0	230.7	granodiorite
			209	C	500	12	54.0	60.5	2.6	273.4	granodiorite
			215	A	400	12	56.3	46.8	2.0	463.8	granodiorite
			219	C	500	10	−116.4	−43.4	5.0	95.4	granodiorite
Sukurae Group	Nakano Fm N4		216*	C	500	9	−115.8	−56.0	18.9	8.4	dacite welded tuff
	N3	63	106*	A	500	12	−100.2	−27.0	8.9	22.5	dacite welded tuff
	N2	58	108*	B	500	8	−107.7	−15.8	16.3	12.5	dacite welded tuff
			218*	A	500	11	98.8	10.9	3.4	181.1	dacite welded tuff
	N1		109*	A	400	10	−64.4	−72.9	2.7	325.3	rhyodacite lava
Ochi Group	Yadani Fm.	80 < < 92	111*	C	500	11	−131.5	−44.7	6.7	47.1	rhyolite tuff
			111'*	B	500	6	−91.1	−36.4	13.5	25.6	mud stone
			205*	C	500	12	−119.7	−63.1	2.2	377.2	rhyolite welded tuff
	Ichiibara Fm.	92	112*	C	500	8	−72.7	−53.4	4.9	130.6	rhyolite welded tuff
			206*	C	500	11	−137.4	−35.8	6.4	51.5	rhyolite welded tuff
			1137*	B	550	9	−99.3	−18.2	9.7	29.0	rhyolite welded tuff

Data are after tilting correction. Asterisk (*) denotes a site where dip of the strata were observed. Kawamoto Granodiorite is estimated to be subjected to little regional movement. The ages are fission-track data after Matsuda [8] ([a] age estimated based on paleontology [13,14]; [b] K-Ar data from Kawano and Ueda [9]). Category: behavior of NRM direction during demagnetization (see text). DT is the magnetic cleaning temperature, N is the number of samples in site mean computation, D is the site mean declination, I is the site mean inclination, and α_{95} and k are Fisher statistic parameters.

of NRM was completely obscured by the secondary component which was acquired after the Middle Miocene. There remains, however, another possibility for the explanation of the directions during this period, which is that the declination change began at about 23 Ma and finished within one million year. Further paleomagnetic and chronological work is in progress and will hopefully enable us to refine the time of the declination change. The data from the four sites formed during the period of the declination change were excluded from the statistical computation of the Miocene.

4. Discussion

A statistical analysis [10] is given for each stage in Table 2, together with virtual geomagnetic poles.

Comparing the pole position of the Cretaceous with previously reported data from Southwest Japan [15], the circles of 95% confidence overlap. This indicates that the study area appropriately represents the tectonic block of Southwest Japan since the Cretaceous. Both declination values of Cretaceous and Paleocene age are relatively larger than the previous data which were obtained without intensive thermal demagnetization [1,5]. The more easterly direction in this study can be attributed to successful removal of a secondary component by thermal treatment. The secondary component remaining without demagnetization generally makes the remanent magnetization direction fall closer to that of the present field. The inclination value of the Cretaceous samples seems to be shallower than previous data [15]. Since the direction in this study is fairly scattered ($\alpha_{95} = 20.3°$), it is difficult to postulate the northward translation of this area at 95% confidence level. The shallow inclination and high dispersion of data are probably due to either the uncertainty of the tilting correction or the masking of the primary direction by some hard secondary components.

The change in declination is plotted as a function of the age determined by fission-track dating in Fig. 5. The easterly declination value decreases gradually from the Late Cretaceous of 92 Ma to the Late Oligocene of 28 Ma. While the easterly declination persists until 28 Ma, the declination at about 12 Ma coincides with that expected from an axial dipole. This clearly indicates that the main part of the declination change began after 28 Ma.

The declination change is caused both by tectonic rotation and by apparent polar wander. The tectonic rotation of Southwest Japan is calculated through comparison between declinations of Southwest Japan and the Korean Peninsula, since the paleomagnetic data from Korean Peninsula only show the geomagnetic effect of apparent polar wander [16]. Differences of declinations between Southwest Japan and the Korean Peninsula were calculated only for the periods of the Paleocene and the Cretaceous, because the paleopole after 30 Ma is consistent with the present geographic north pole [17,18]. The amount of rotation of Southwest Japan is listed in Table 2. The uncertainty in ro-

TABLE 2

Average paleomagnetic directions

Stage	N	$\bar{D}$ (°)	$\bar{I}$ (°)	α_{95} (°)	VGP	∂p	∂m	Δ	Δ_{95}
12 Ma	4	1.3	56.3	16.9	87.9°N, 161.6°E	17.6	24.4	1	32
28–33 Ma	11	58.0	43.0	10.4	39.5°N, 142.8°W	8.0	12.9	58	14
59–63 Ma	6	80.9	42.2	25.6	20.8°N, 153.4°W	19.3	31.5	54	40*
80–92 Ma	6	71.3	44.2	20.3	29.1°N, 150.3°W	16.0	25.5	45	34*

N is the number of sites. $\bar{D}$, $\bar{I}$ are mean declination and inclination for each period, α_{95} is circle of 95% confidence. Δ is the difference in declination value between this area and Korean Peninsula. Data for Korean Peninsula from Otofuji et al. [16]: $D = 26.6°$, $I = 62.3°$ and $\alpha_{95} = 8.3°$. Δ_{95} is uncertainty in declination at 95% confidence, where $\Delta_{95} = \sin^{-1}(\sin \alpha_{95}/\cos I)$. Uncertainty of older period (*) is calculated as follows: $\sqrt{(\Delta_{95})^2_{\text{Korea}} + (\Delta_{95})^2_{\text{this area}}}$.

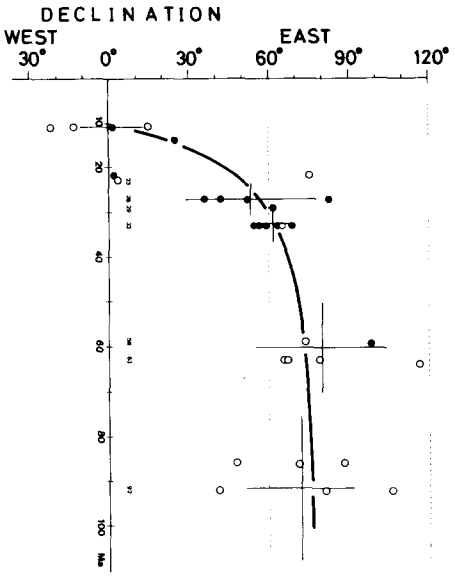


Fig. 5. Declination data of site means as a function of fission-track age together with representative mean directions for five periods. The declination error bars are the α_{95} values and age error bars are the uncertainty of the fission track method of 20%. Closed circles show the data with normal polarity, open circles those with reversed polarity. The solid curve is presumable declination change.

tation at the 95% confidence level is the Δ_{95} value after Kellogg and Reynolds [19] where $\Delta_{95} = \sin^{-1}(\sin \alpha_{95}/\cos I)$ with α_{95} the radius of the 95% confidence circle and I the mean inclination. The amount of rotation scarcely decreases with decreasing age between 92 and 28 Ma, although the uncertainty is quite large. The declination change during the Paleogene probably results from the effect of polar wander. The results suggest that the tectonic rotation of Southwest Japan took place abruptly after 28 Ma.

The amount of rotation of Southwest Japan was estimated from the data of the Kawamoto Granodiorite and the Sasahara Formation with fission-track ages between 28 and 33 Ma, because these directions scarcely include the effect of the polar wander motion, and the geomagnetic secular variation would be eliminated from their mean direction by averaging over enough time. It is concluded that Southwest Japan has been subjected to a clockwise rotation through $58 \pm 14°$. Inclination data indicate that Southwest Japan has not undergone significant north-south translation.

The reliability of the timing of 28 Ma is ascertained through the concordance of the fission-track data within same rock unit and comparison of fission-track ages with one determined by the K-Ar method. The fission-track ages of 29 Ma and 33 Ma are obtained from the Kawamoto Granodiorite which is overlain by the Sasahara Formation with a fission-track age of 28 Ma. These ages show good agreement with a K-Ar age (25 Ma) determined on biotite [9], although the K-Ar age is relatively younger than the other ages. The younger age of K-Ar may be attributed to the loss of argon from biotite because of alteration of rock samples from the Kawamoto Granodiorite [9].

5. Conclusion and implications

The paleomagnetic results from Upper Cretaceous to Miocene rocks from the Gō River region established: (1) the clockwise rotation of $58 \pm 14°$ of Southwest Japan took place later than 28 Ma; (2) the rotational motion terminated around 12 Ma; and (3) Southwest Japan was subjected to little rotational motion between 92 and 28 Ma.

A possible 28 Ma paleogeographic reconstruction is proposed (Fig. 6). Since the amount of clockwise rotation of 58° is 1.8 times as great as that previously reported [5], this puts a strong constraint on the position of the rotation pole. Southwest Japan can be rotated without collision with neighbouring regions through 58° only about the pivot within the shaded zone of Fig. 6a. To obtain the paleoposition, Southwest Japan was rotated through 58° about a pivot around 34°N, 129°E in this zone (Fig. 6b) based on the following assumptions: (a) the rotational motion of Southwest Japan is associated with the opening of the Japan Sea [20–22], and (b) the pivot of rotation is located near the end of the moving block, which is suggested from studies on western North America [23–25]. Our model is similar to that of Honza [26] which was postulated from the viewpoint of the configuration of oceanic and continental crust around the Japan Sea.

Our 28-Ma reconstruction is examined through

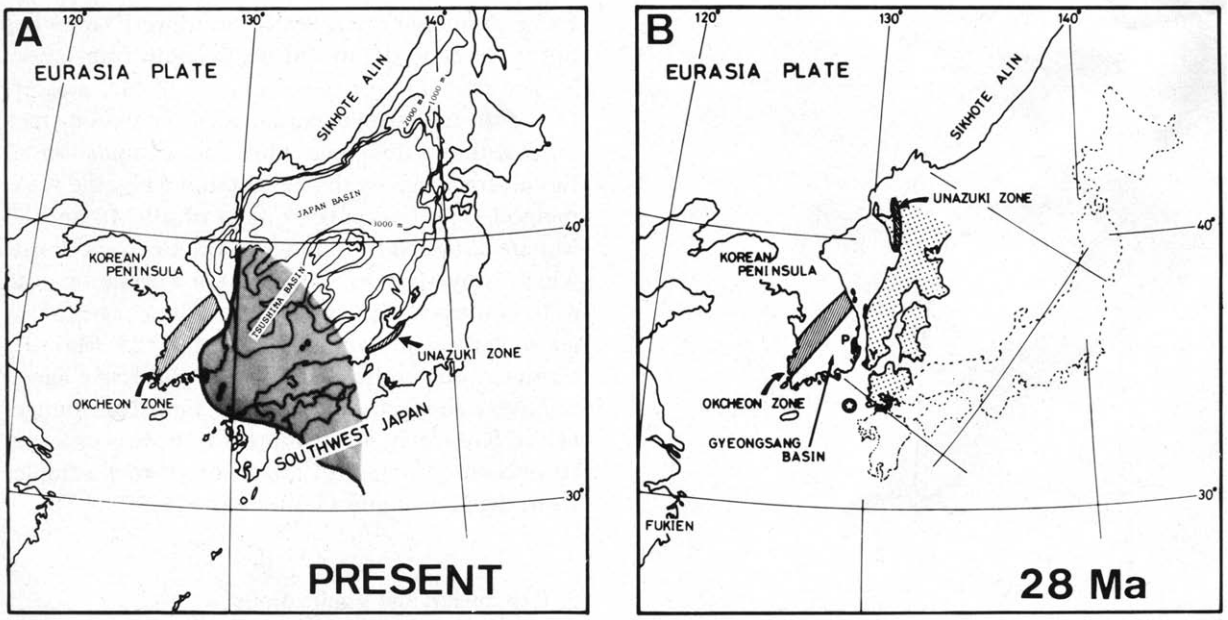


Fig. 6. The rotation of Southwest Japan. A. Location of pivot of rotation of Southwest Japan. Location of pivot is limited to the shaded zone. Southwest Japan can be rotated without collision and overlap with the Korean Peninsula or Siberia through 58° about a pivot in this zone. Hachures indicate the Okcheon and Unazuki zones which are interpreted to be a continuous geotectonic unit [35]. B. Paleoposition at 28 Ma, before beginning of rotation. The pivot of rotation is shown by a star at the western end of Southwest Japan at 34°N, 129°E. Black areas are outcrops of Tertiary sediments of the Pohang area of the Korean Peninsula and the Yamaguchi area of Southwest Japan. P = Pahong area; Y = Yamaguchi area.

the correlation between the geological evidences from Southwest Japan and the Korean Peninsula, since the Korean Peninsula likely has clues to the paleoposition of Japan as originally suggested by Carey [27]. Cretaceous to Paleogene granites in Southwest Japan correlate with the numerous granites intruded into the Gyeongsang Basin of the Korean Peninsula [9,28,29]. The Mesozoic granites of Southwest Japan may form a part of the long belt of the Mesozoic granites [30] traced from the Fukien of southern China, through the Gyeongsang Basin, to Sikhote Alin. The Unazuki metamorphic belt of Southwest Japan can be located in the northeast prolongation of the Okcheon zone of Korea when this reconstruction is made. The match between the Tertiary outcrops in the Pohang area of Korea and the Yamaguchi area of Southwest Japan [31,32] also results from the reconstruction. Our reconstruction leads to a good correlation. Timing of the opening of the Tsushima Basin is confirmed to be later than 25 Ma by the paleontological data obtained from five deep wells in the Tsushima Basin [33,34], supporting the hypothesis that Southwest Japan flanked on the Korean Peninsula at 28 Ma.

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